Lab Procedure
Camera

Introduction

Ensure the following:

1. You have reviewed the Application Guide – Camera
2. Make sure the **Mechatronic Sensors Trainer** is out of its case; it is powered using the provided USB cable connected to the PC.
    a. The LED next to the USB C port will stay white after the touch screen starts loading.
    b. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.

Exploring Camera Sensors

1. Open the provided OV5640_datasheet.pdf located in (…/sensors_trainer/1_fundamentals/camera) and look for the following:
    a. Pixel size (value in µmm)
    b. Shutter type
2. Open the provided (name of lens spec sheet) and look for the following
    a. Diagonal FOV
3. Launch Visual Studio Code or your preferred Python IDE. If using Visual Studio Code, click on **File** > **Open Folder**…, and browse to the directory containing the python files and lab procedure for the Camera lab (…/sensors_trainer/1_fundamentals/camera/hardware/python). Open this directory.
4. Open **camera_display.py** and modify the parameters:
    a. `frameRate` (Example frame rate values: 10,15,30)
    b. `imageSize` (Example resolutions [160,120], [640,480], [1280,720], [1920,1080])
5. Move the camera around and examine the corners, is there distortion? What does this tell you about camera sensors? As you change framerates and resolutions, are all resolutions supported by all framerates? If you move an object quickly, is there an effect on the video display?
6. Press `Enter` on your keyboard once the example is finished.

7. Open camera_calibration.py

8. The next step will require a physical calibration board. You can use the following - link to download a version of a checkerboard for an 7x9 board on A4 paper with a 20mm square. The same checkerboard is also available in the camera folder. Make sure you place the 7x9 board on a flat surface, you can tape it to keep it from bending. We recommend to the back of a hardcover book or a piece of cardboard.

9. Change the parameter **`calibrate`** to **`True`**.

10. Click ▷ to run the example.

11. Click on the OpenCV window and tap the keyboard key 'q' when you're ready to save a calibration image. Consider:

    a. Keeping the entire chessboard in view

    b. Having different orientations

    c. Having both up close and far away images of the chessboard

12. Once 15 images have been captured the camera calibration sequence will start. Wait until the camera and distortion parameters have been calculated. Make sure your camera calibration/projection error is <1. This value is in pixels and the closer to 0 it is, the better.

13. What is the focal point of the camera? The manufacturer provided the lens focal length as 2.6mm. Using the equation:

$$f_{mm} = f_{pxl}\, p_{xl} \tag{1}$$

Note: f_{mm} is the focal length in mm, f_{pxl} is the focal length in pixels and $p_{xl}$is the pixel size in µm.

How do calibrated focal length and manufacturer provided focal length vary?

14. Using the calibration parameters for the sensor image, calculate the diagonal FOV. How does the estimated and actual FOV vary? Use the following equation:

$$FOV_{diag} = \arctan\left(\frac{\sqrt{w^2 + h^2}}{2f_{pxl}}\right) \tag{2}$$

Note: w and h are the width and height of the image in pixels.

15. Make a note of the **Mechatronic Sensors Trainer** serial number, resolution, camera intrinsic and distortion parameters. They are needed for when the camera needs to be calibrated for more specialized use cases.

16. Up until this point we have looked at the camera in grayscale to understand the sensor specific parameters, modern cameras also provide color information. Open color_detection.py

17. Open the concept review for color spaces found under (…/4_concept_reviews/Perception) to review how colors are represented in an image.

18. In this example we use computer vision techniques to select a unique color range seen in a camera. Change the array **`bgrValuesLower`** and **`bgrValuesUpper`** to define a window of allowable colors. Values can vary between 0-255.

    As an example, the range:

    ```
    bgrValuesLower = [0,0,100]
    ```

    ```
    bgrValuesUpper = [50,50,255]
    ```

    Will detect colors which contain primarily red as well as some blue and green.

    Note: In OpenCV the color vector is defined as [Blue, Green, Red]

19. Click  to run the example.

20. Change the environment (close a window, dim lights or add another light source). Were you still able to detect the color in the camera image?

21. Stop, save and close your program.

22. Power OFF the Mechatronic Sensors Trainer by disconnecting its USB C cable, disconnect any external sensors and pack them along the base and the USB cables in the provided case.